

THEORY



Training Program 01

Advanced NLP & Large Language Models (LLMs)



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Training Program 01

M01: Text Preprocessing

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects



MODULE M01

TEXT PREPROCESSING

CHAPTER M06

Multimodal NLP



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6.1 Introduction to Multimodal and Cross-Modal NLP

6.2 CLIP: Bridging Vision and Language

6.3 Whisper: Unified Speech Processing (ASR)

6.4 Knowledge-Augmented Text Enrichment

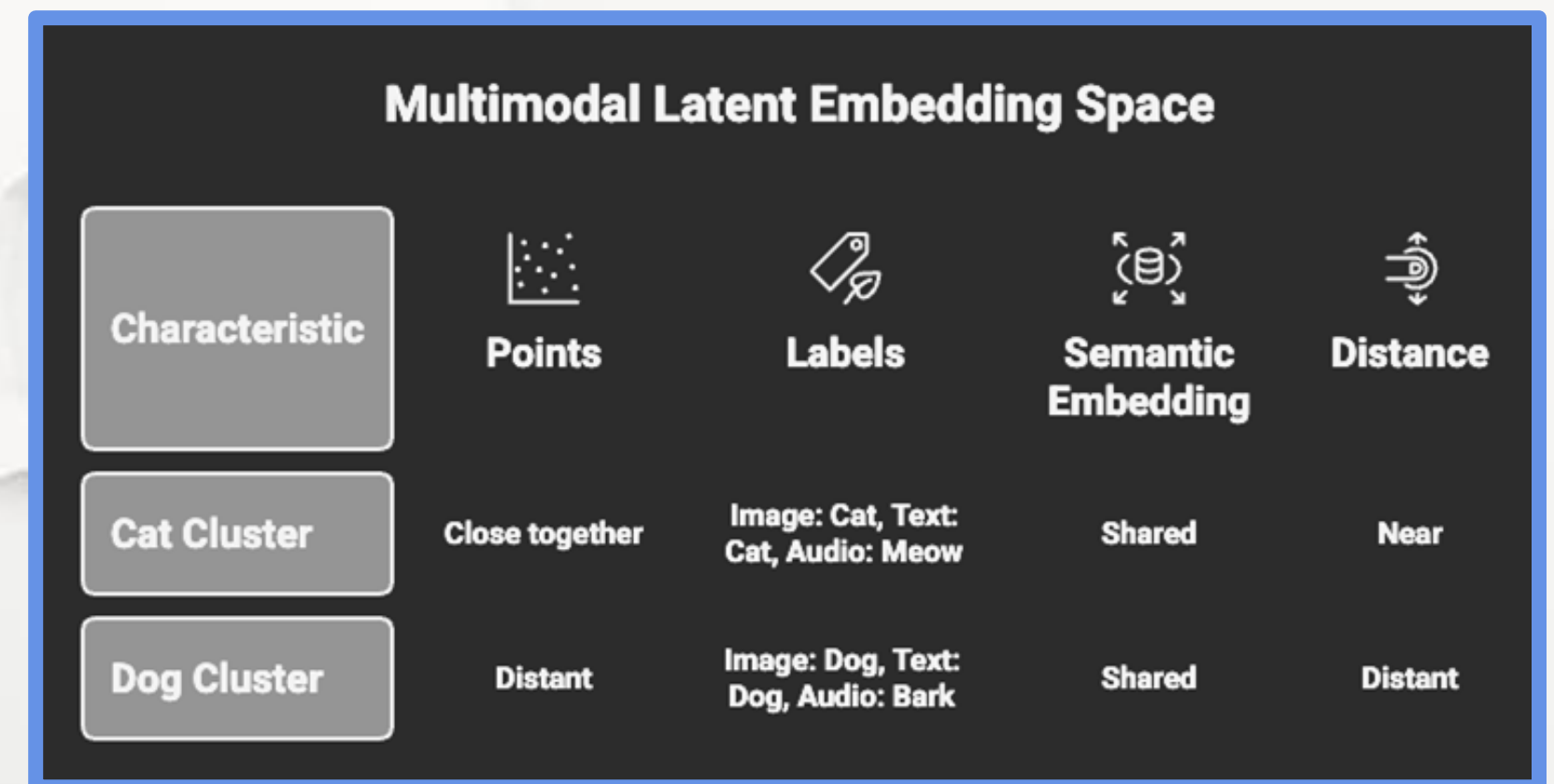
6.5 Summary and the Future of Unified AI



Chapter 6.1 Introduction to Multimodal and Cross-Modal NLP

Multimodal vs Cross-Modal AI

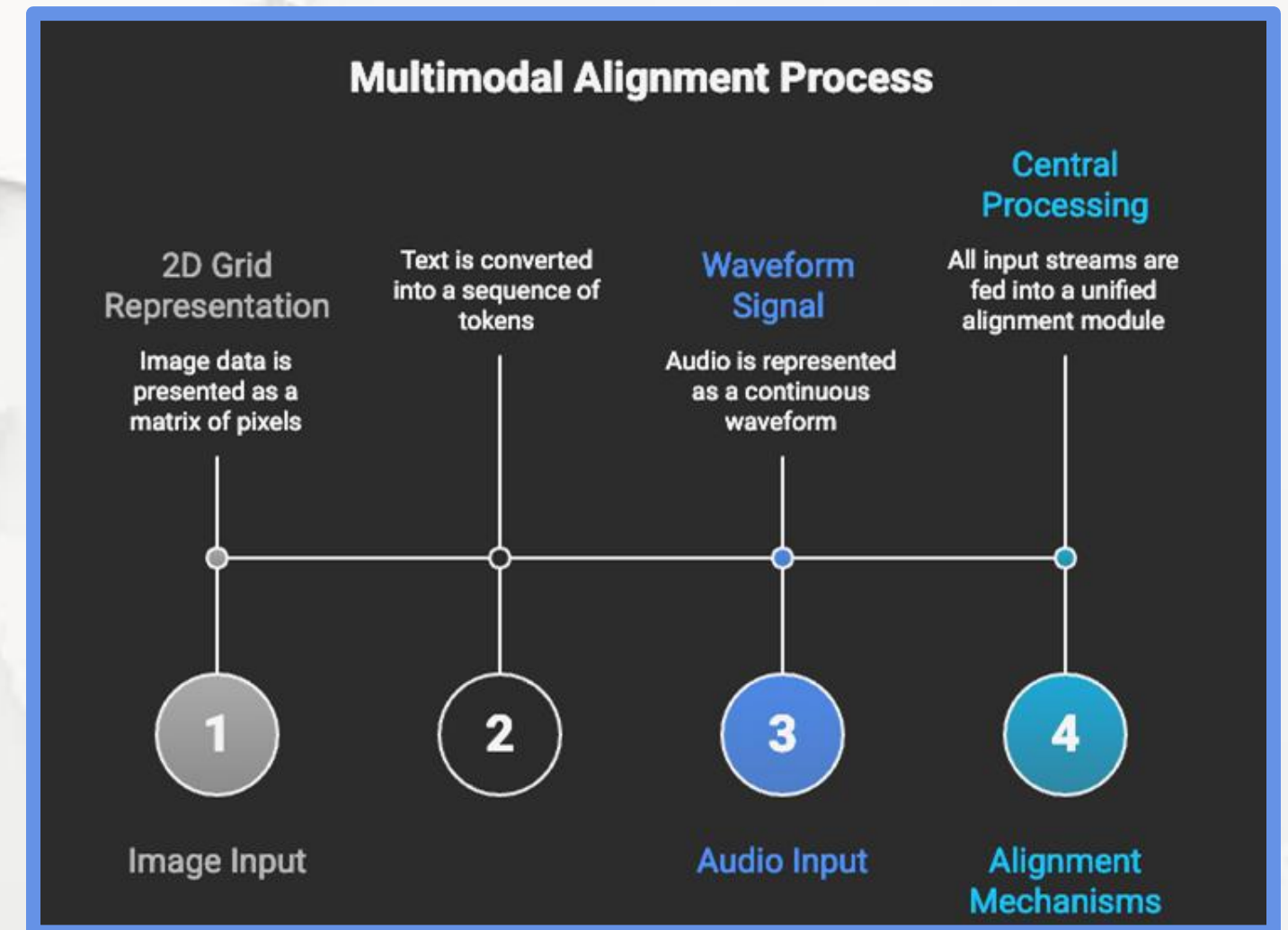
- Multimodal AI: simultaneous processing of multiple modalities.
- Cross-modal AI: translation between modalities (text $\leftrightarrow$ image $\leftrightarrow$ audio).
- Core challenge: semantic alignment across heterogeneous data.
- Unified latent space enables modality-agnostic reasoning.



Chapter 6.1 Introduction to Multimodal and Cross-Modal NLP

The Alignment Problem & Unified Representations

- Modalities differ in structure, scale, and noise characteristics.
- Alignment achieved via contrastive learning and cross-attention.
- Semantic similarity replaces symbolic matching.
- Foundation of generalizable multimodal intelligence.



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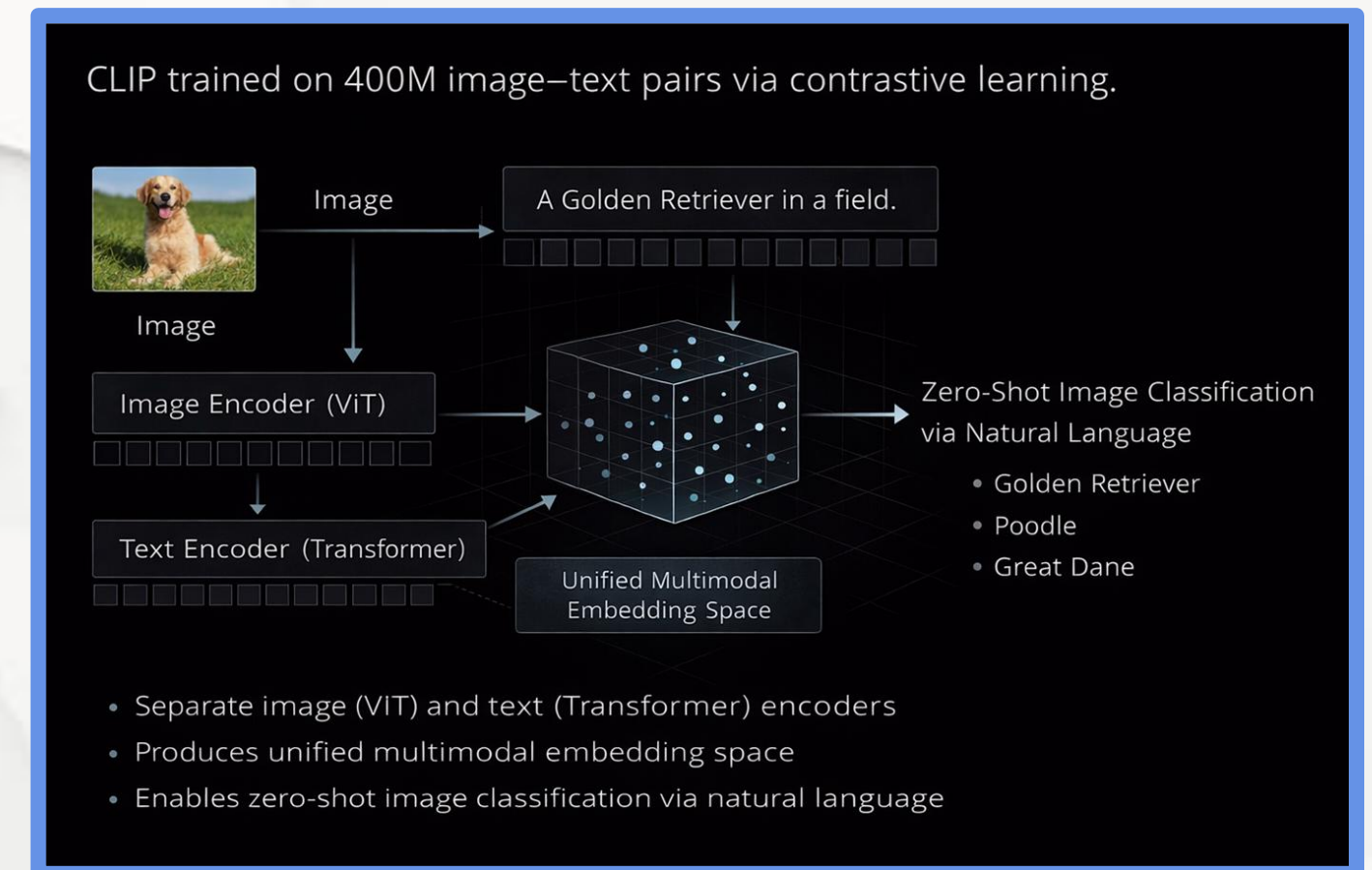
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Chapter 6.2 CLIP: Bridging Vision and Language

Vision–Language Bridging

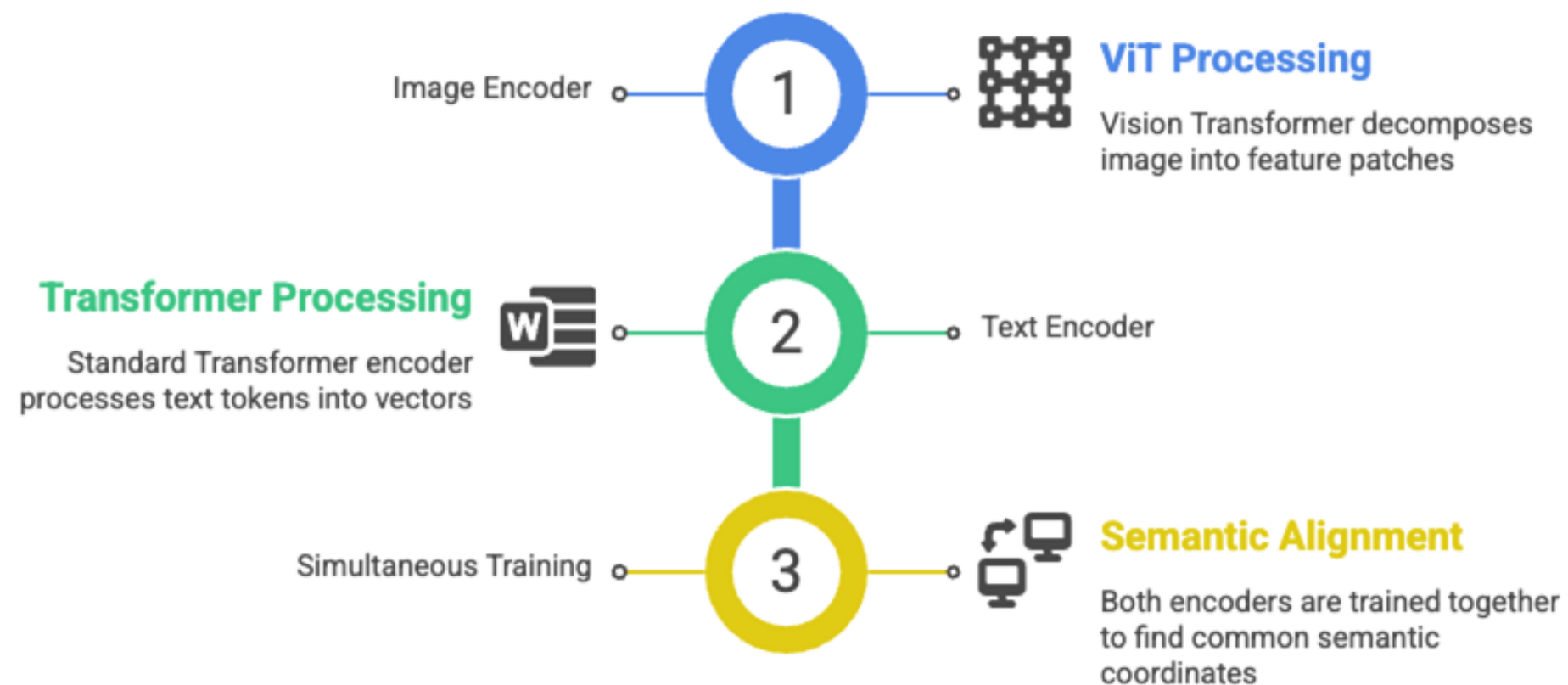
- CLIP trained on 400M image–text pairs via contrastive learning.
- Separate image (ViT) and text (Transformer) encoders.
- Produces unified multimodal embedding space.
- Enables zero-shot image classification via natural language.



Chapter 6.2 CLIP: Bridging Vision and Language

Separate Text and Image Encoders

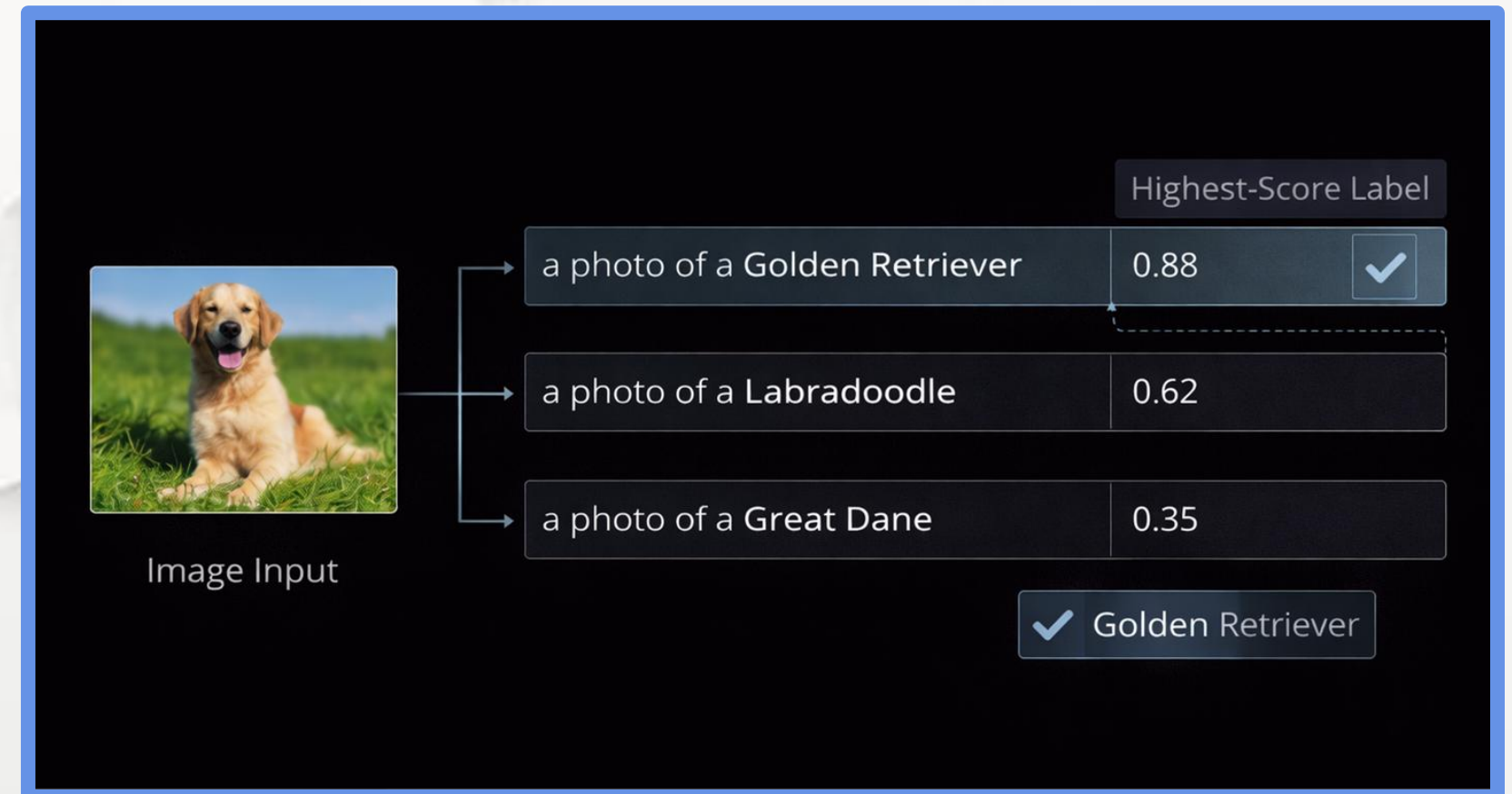
CLIP's Parallel Encoder Architecture



Chapter 6.2 CLIP: Bridging Vision and Language

CLIP in Practice: Zero-Shot & Generative Guidance

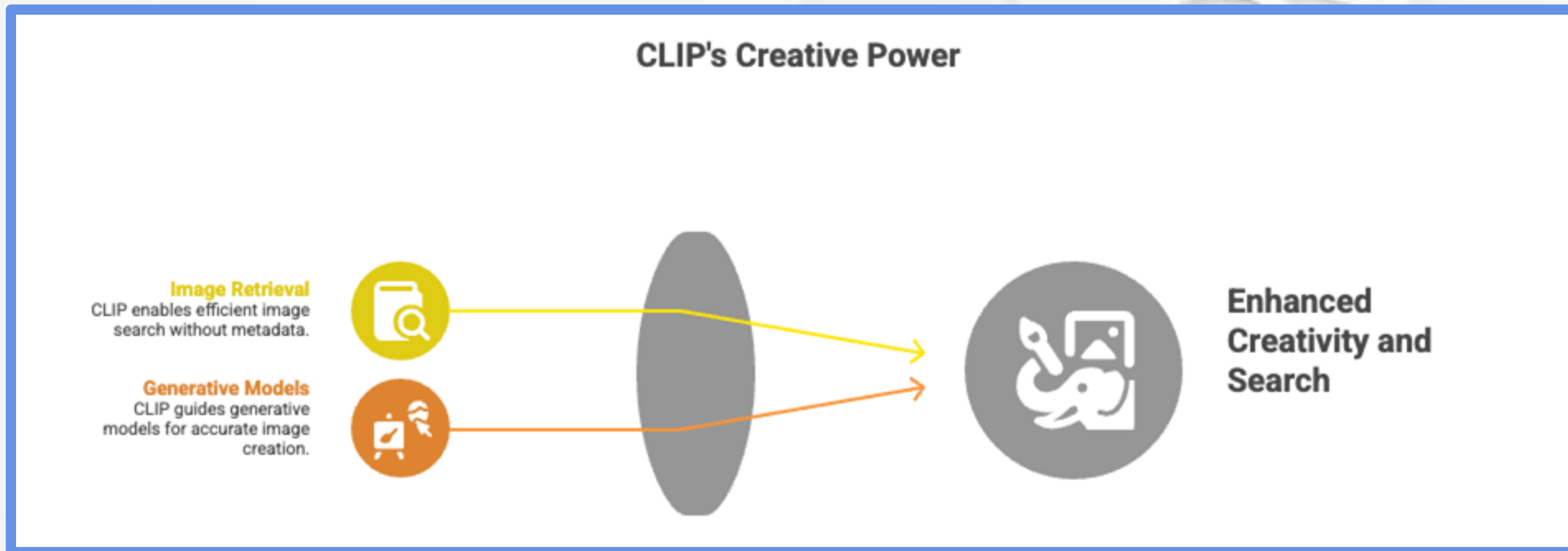
- Zero-shot image classification without task-specific training
- Natural language prompts act as dynamic labels
- Backbone for image retrieval at scale
- Semantic guidance engine for text-to-image generation



M06

Chapter 6.2 CLIP: Bridging Vision and Language

Applications: Image Retrieval and Captioning



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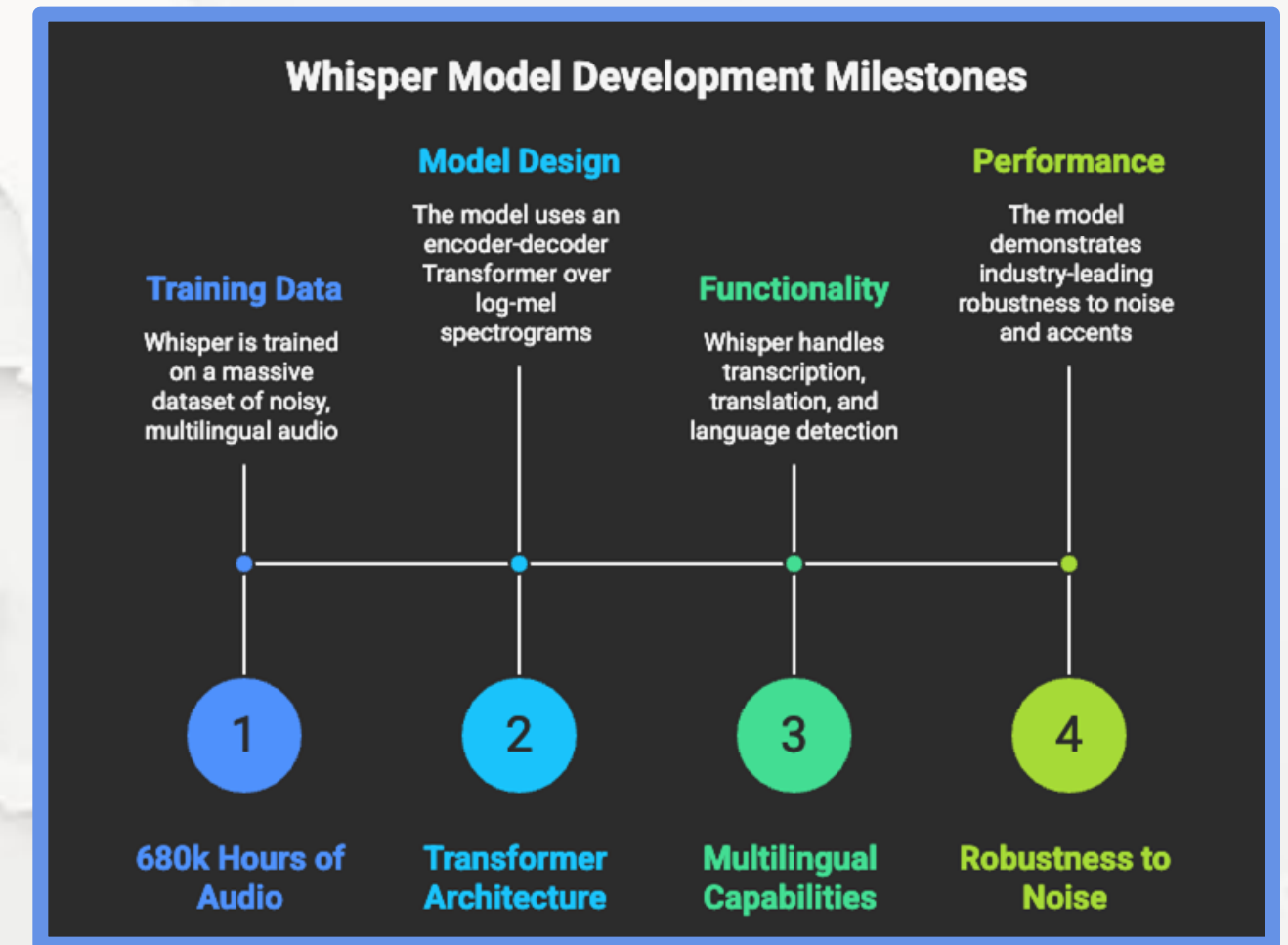
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Chapter 6.3 Whisper: Unified Speech Processing (ASR)

Whisper: Robust Speech-to-Text

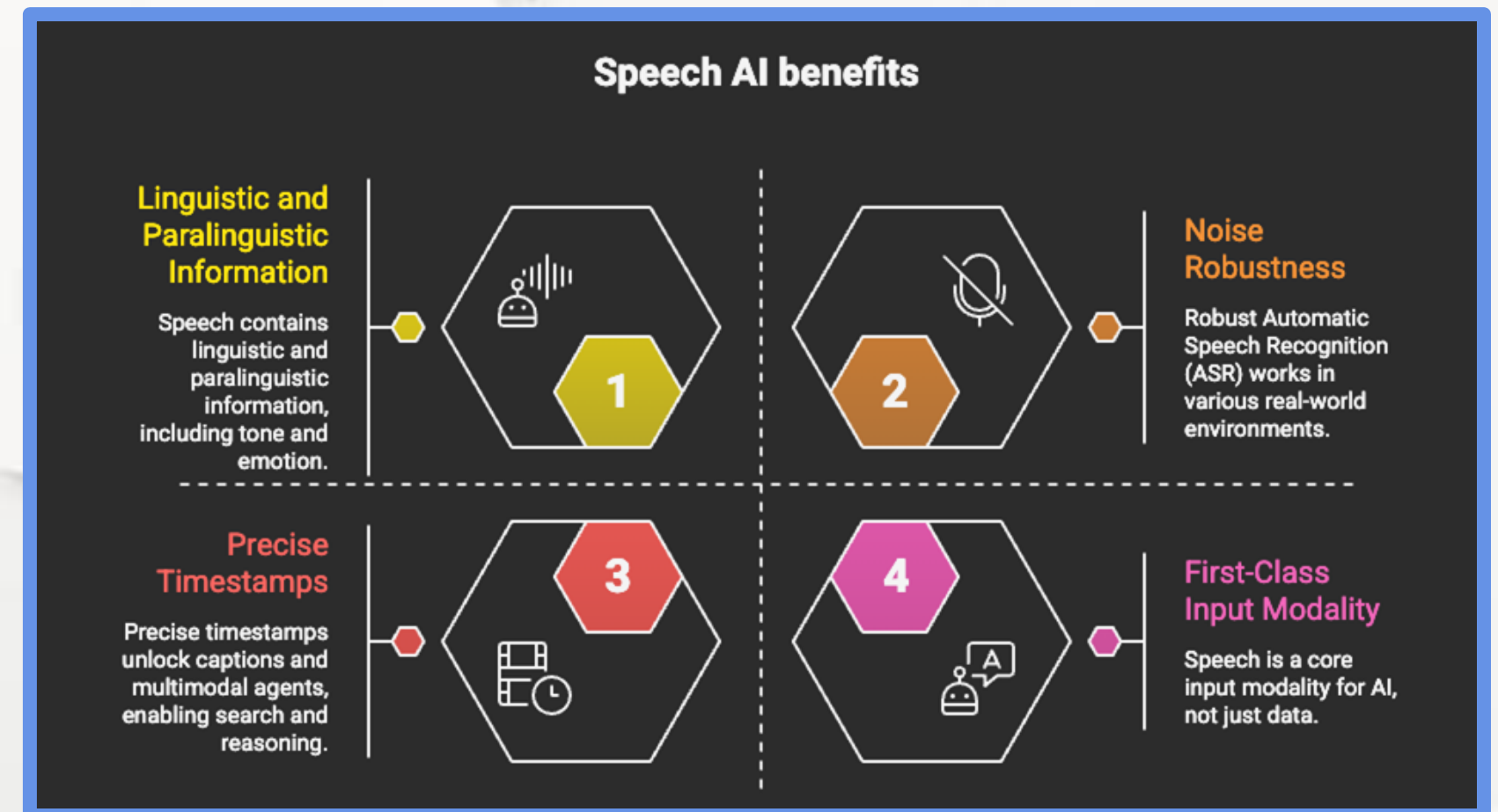
- Whisper trained on 680k hours of noisy, multilingual audio.
- Encoder-decoder Transformer over log-mel spectrograms.
- Handles transcription, translation, and language detection.
- Industry-leading robustness to noise and accents.



Chapter 6.3 Whisper: Unified Speech Processing (ASR)

Speech as a Multimodal Signal

- Speech contains linguistic + paralinguistic information
- Noise robustness enables real-world deployment
- Precise timestamps unlock captions and multimodal agents
- Speech becomes first-class AI input modality



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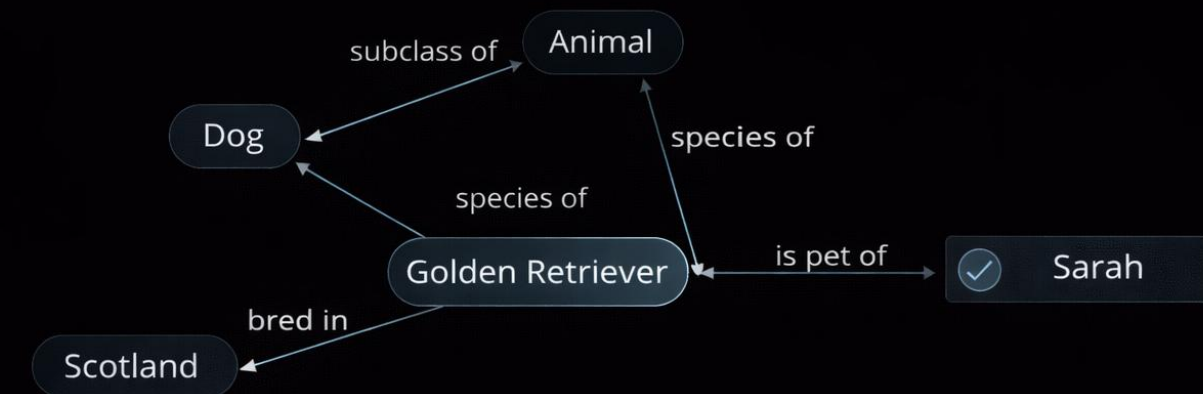


Chapter 6.4 Knowledge-Augmented Text Enrichment

Knowledge Graphs as the Anchor of Truth

- Knowledge Graphs represent facts as entities and relations
- Triples (subject–predicate–object) form deterministic truth layer
- Neural models lack factual persistence without KGs
- KGs enable explainability and verification

Semantic Knowledge Graph



Example Triples:

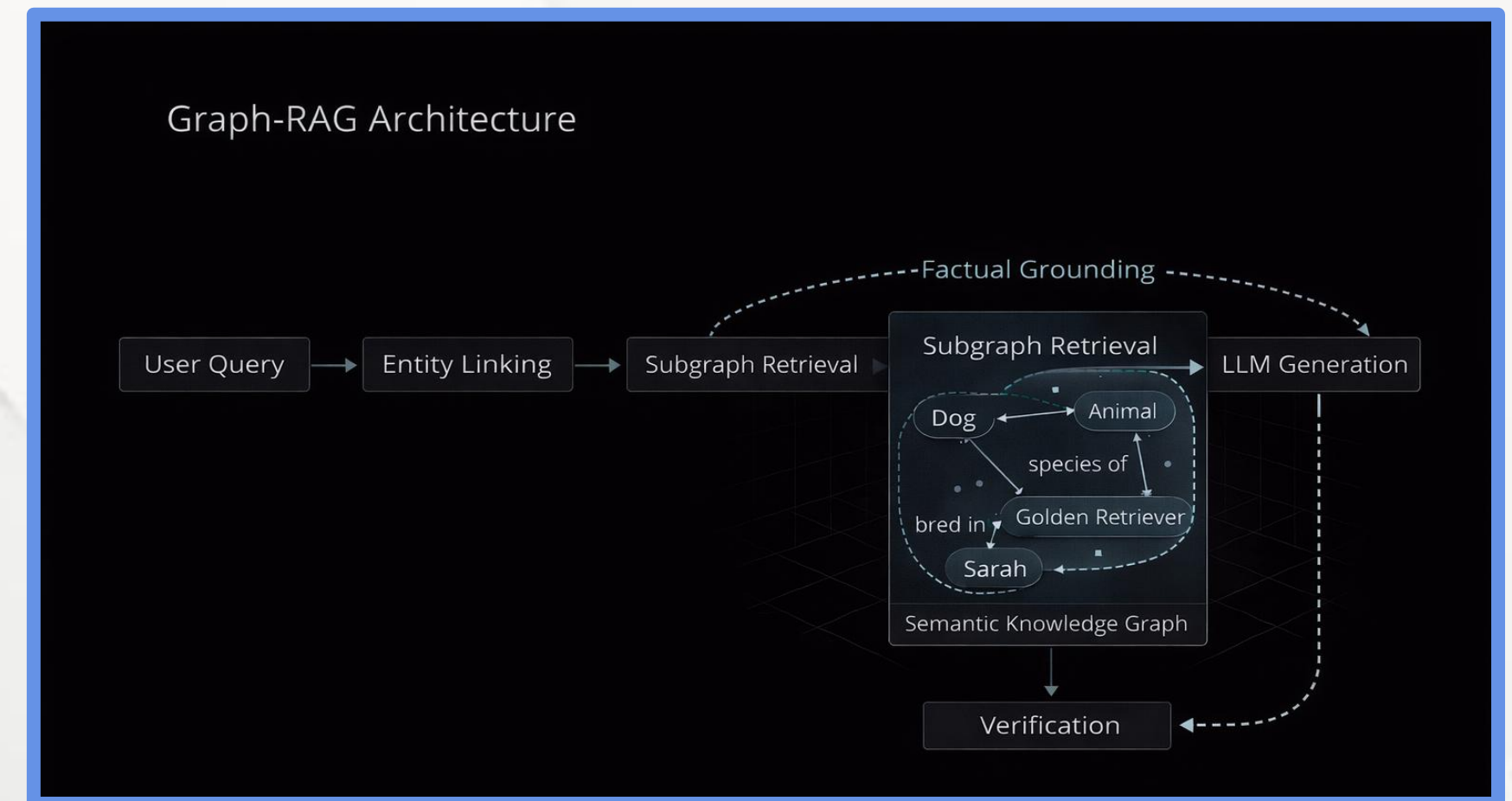
- Golden Retriever → species of → Dog
- Golden Retriever → is pet of → Sarah
- Sarah → lives in → New York City



Chapter 6.4 Knowledge-Augmented Text Enrichment

Knowledge-Augmented Generation & Graph-RAG

- KG embeddings bridge symbolic graphs and neural vectors
- Graph retrieval replaces raw text retrieval
- Sub-graph grounding reduces hallucinations
- Gold standard for regulated domains (law, medicine)



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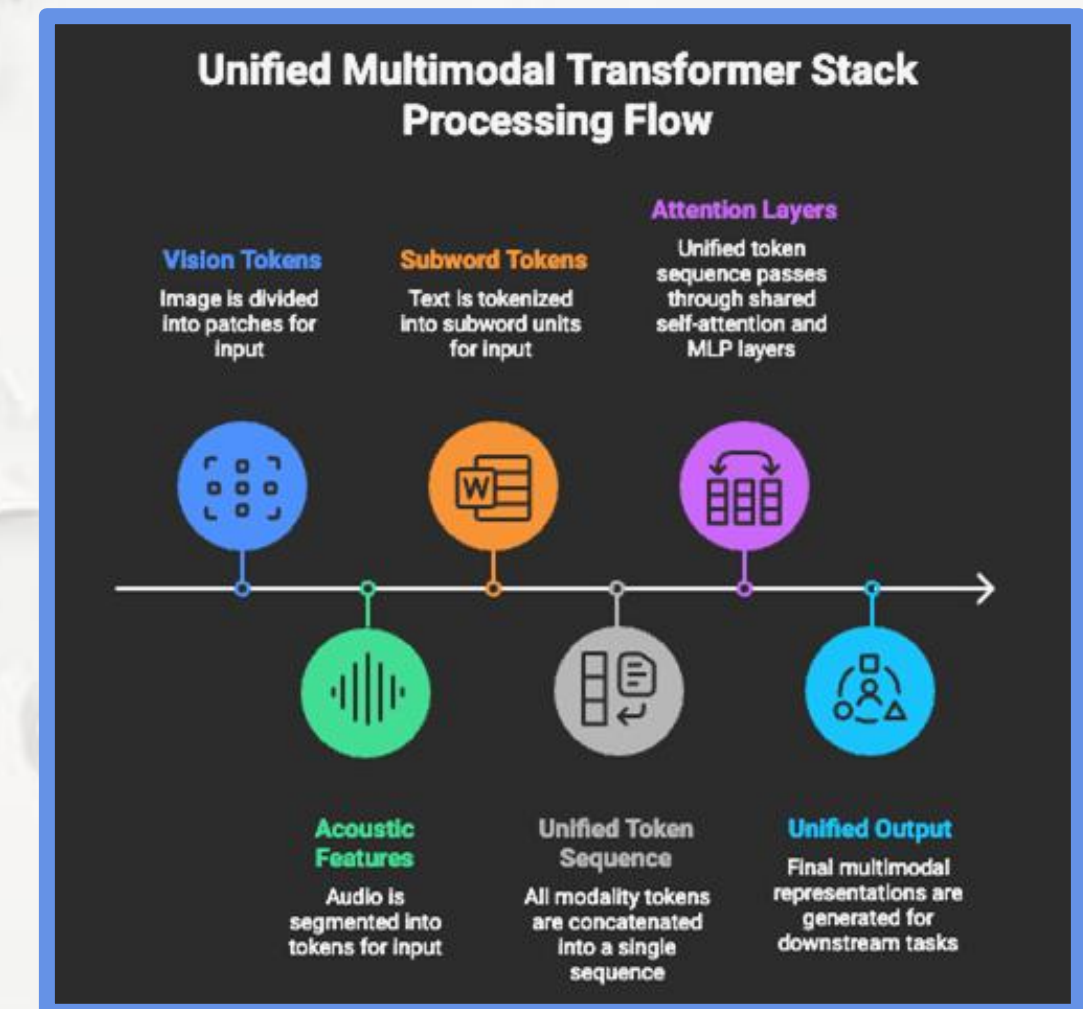
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From Modular Pipelines to Unified Models

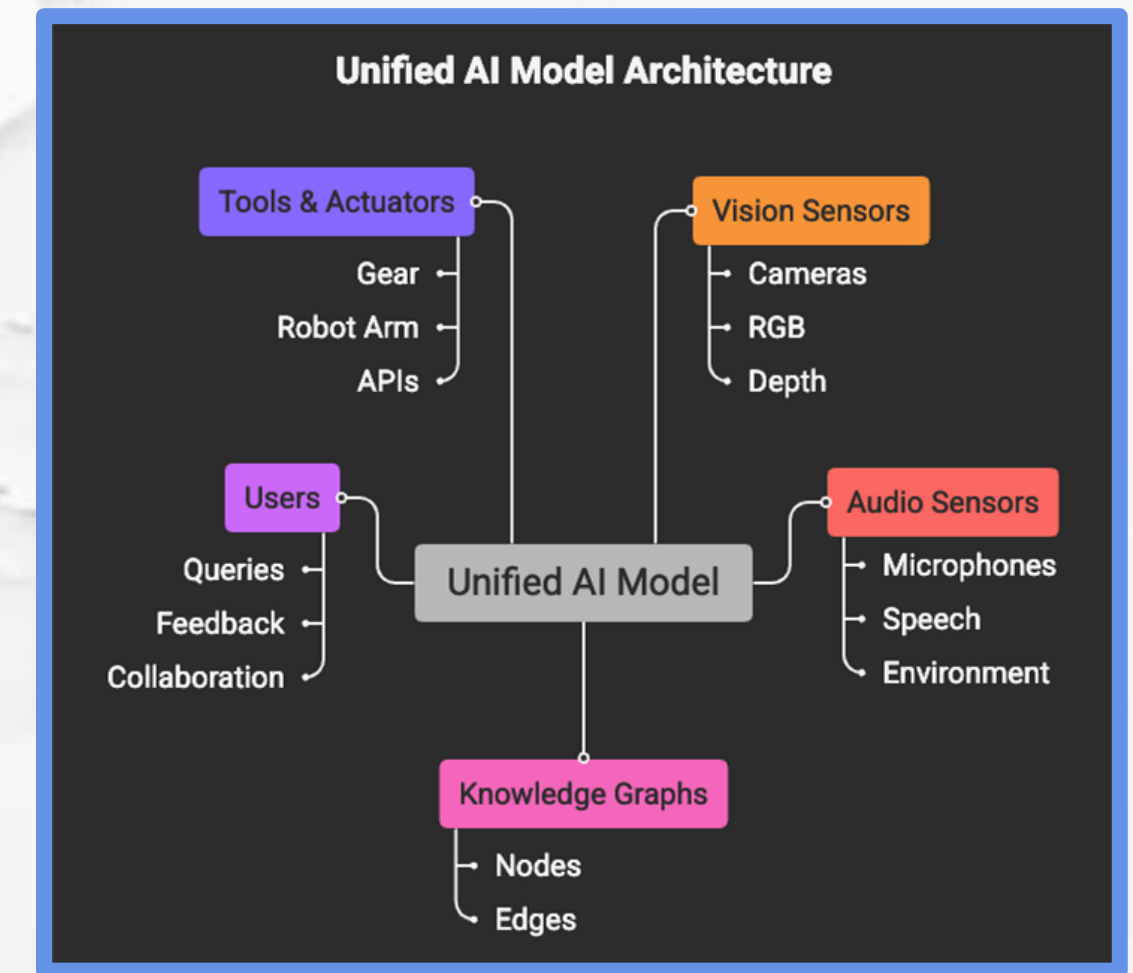
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Risks, Ethics & Bias in Multimodal AI

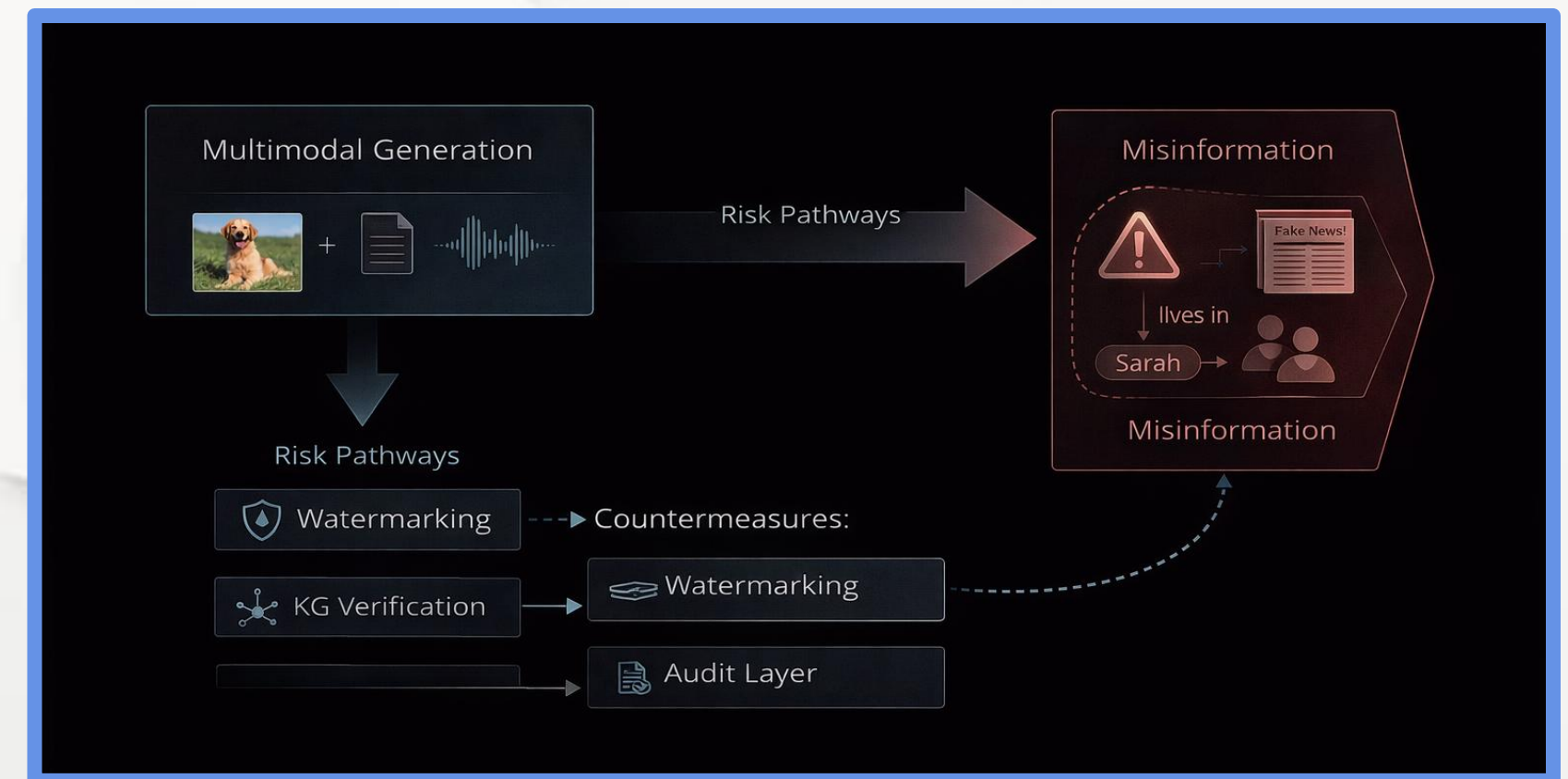
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Chapter 6.5 Summary and the Future of Unified AI

The Future: Unified AI Systems

- Language as cognitive core, multimodality as sensory system.
- Alignment across modalities enables real-world intelligence.
- Knowledge grounding ensures reliability.
- Foundation for agentic, autonomous systems.



Academic & Technical References

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Academic & Technical References

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